Refik Anadol: In-Depth Research Report

Data as pigment, machine dreams as canvas

2026-10-15

- Refik Anadol: In-Depth Research Report
  - 1. Summary
  - 2. Life and formation
    - Intellectual starting points
  - 3. Major works by period
    - 3.1 Data painting on architecture (2008 to 2015)
    - 3.2 Machine hallucinations (2016 to 2023)
    - 3.3 In-house models based on nature and architecture (2024 to present)
  - 4. Case study: *Unsupervised* (MoMA, 2022 to 2023)
  - 5. DATALAND (2026)
  - 6. Visual grammar: recurring vocabulary
  - 7. Criticism and debate
  - 8. Awards (selected)
  - 9. Implications for implementation
- Refik Anadol Timeline
  - Overview
  - Evolution of technique
  - Major exhibition periods, 2022 to 2026
- Technique Breakdown: Refik Anadol's Data Aesthetics Pipeline
  - 1. Overall pipeline
  - 2. Stage by stage
    - 2.1 Collection: from "public data" to "permission-based data"
    - 2.2 Training and embedding
    - 2.3 Latent space exploration
    - 2.4 Real-time rendering
    - 2.5 Architectural display
  - 3. Mapping to the web (three.js)
- Scene Catalog: 10 Themes × 10 = 100

    - 1 Machine Dreams
    - 2 Atmosphere
    - 3 Ocean & Water
    - 4 Forest & Life
    - 5 Earth & Geology
    - 6 Cosmos
    - 7 Micro
    - 8 City
    - 9 Countryside
    - 10 Physics & Math
- Sources
    - Primary sources (artist, institutions)
    - Secondary sources (encyclopedia, press, criticism)
    - Conflicts between sources

Refik Anadol: In-Depth Research Report

> Research date: 2026-10-15. “Refik Anotol” in the original request was interpreted as the Turkish American media artist **Refik Anadol**. Figures and dates follow the sources in `sources.md`; where sources disagree, both values are given in the text.

1. Summary

Refik Anadol (born November 7, 1985, in Istanbul) is a media artist who works with “data as pigment and machine intelligence as collaborator.” In 2014 he founded Refik Anadol Studio (RAS) in Los Angeles with Efsun Erkılıç, and on June 20, 2026, he opened DATALAND, which he calls “the world’s first museum of AI arts,” at The Grand LA in downtown Los Angeles.

His work falls into three broad periods.

1. **Data painting on architecture (2008 to 2015)**: projection mapping turns building surfaces and whole rooms into moving canvases. Key works: *Infinity Room* and *Visions of America: Amériques*.
2. **Machine hallucinations (2016 to 2023)**: after the Google AMI residency, he visualizes the latent space of huge archives with GANs and dimensionality reduction (t-SNE, UMAP). This period includes *Archive Dreaming*, *WDCH Dreams*, the *Machine Hallucinations* series, *Unsupervised* at MoMA, and the exterior of the Las Vegas Sphere.

3. **In-house models based on nature and architecture (2024 to present)**: the Large Nature Model (LNM), trained on permission-based nature data; the Large Architecture Model (LAM), fine-tuned on architectural archives; and DATALAND, which makes them permanent.

Opinion is sharply divided. MoMA added *Unsupervised* to its collection (its first generative artwork), and visitors stayed an average of 38 minutes. The critic Jerry Saltz, by contrast, dismissed the same work as "a massive techno lava lamp" and "a half-million-dollar screensaver." This tension is itself central to understanding Anadol's work.

2. Life and formation

Item	Details
Birth	1985-11-07, Istanbul, Türkiye. Family of teachers
Childhood turning points	Taught himself programming on a Commodore 64 at age 8. The same year he saw *Blade Runner* and was captivated by the replicant whose memories are implanted
Undergraduate	BA in Photography and Video, Istanbul Bilgi University (2009)
Master's 1	MFA in Visual Communication, same university (2011)
Master's 2	MFA in Design Media Arts, UCLA (2014). Mentored by Casey Reas, Jennifer Steinkamp, and Christian Moeller
Teaching	Lecturer, UCLA Design Media Arts
Studio	Refik Anadol Studio (2014, a 1960s warehouse in LA's Frogtown). A team of designers, architects, data scientists, and researchers from 10 countries speaking 12 languages. Runs the research unit RAS LAB
Museum	DATALAND (opened 2026-06-20, The Grand LA)

Intellectual starting points

- **Lev Manovich's "augmented space"**: as an undergraduate he read this paper and was struck by its claim that collaboration between architects and artists could make "the invisible flow of data

visible." He began using the term "data painting" with his first outdoor projection in 2008, and this became his working principle of using "light as material and data as pigment."

- **Blade Runner and memory**: the motif of memories implanted in a machine mind later leads to the core question of Machine Hallucinations: if a machine can learn, it can also remember, dream, and hallucinate.
- **DeepDream (2016)**: Google's release of the DeepDream algorithm during his first Google AMI residency was the direct trigger for that question.

3. Major works by period

3.1 Data painting on architecture (2008 to 2015)

- **Quadrature (2010)**: a live A/V performance with fellow student Alican Aktürk in the main building of santralistanbul. Footage of quadrilaterals was manipulated on the building to explore the relationship between architecture and media. He followed it with *Quadrangle* at the SANAA building in Essen, Germany.
- **Augmented Structures v1.0 (2011, Istanbul Biennial)**: sounds recorded at different times of day on İstiklal Avenue were reinterpreted and projected with nine projectors.
- **Infinity Room (developed at UCLA in 2012, shown at Zorlu PSM for the 2015 Istanbul Biennial)**: a "temporary immersive environment experiment" that uses pixels and 3D projection mapping instead of mirror illusions to turn every surface of a room into an abstract infinite space. It toured to SXSW, the New Zealand Festival, ZKM, and elsewhere.
- **Visions of America: Amériques (2014)**: at the 2013 Microsoft Research Design Expo he proposed using the exterior walls of Walt Disney Concert Hall (WDCH) as a canvas. With the original 3D model from Gehry Technologies, he created an indoor architectural video installation for the LA Phil's performance of Varèse's *Amériques* (his UCLA thesis project).

3.2 Machine hallucinations (2016 to 2023)

- **Wind of Boston: Data Paintings (2017)**: a year of wind data from Boston Logan Airport (speed, direction, gusts, time, and temperature at 20-second intervals) visualized as four "chapters" on a 6 × 13 foot digital canvas. A model early "data painting" that turns the invisible flow of wind into fluid brushstrokes.
- **Archive Dreaming (2017, SALT Galata)**: 1.7 million documents from SALT Research sorted with machine learning. A t-SNE point cloud spreads the whole archive through 3D space, and a DCGAN

generates “alternative documents that never existed.” Visitors explore the archive inside a responsive environment.

- **Melting Memories (2018, Pilevneli Gallery, February 7 to March 17)**: prompted by his uncle’s Alzheimer’s diagnosis, he worked with UCSF’s Neuroscape lab. Anonymous EEG data drives visualizations of memory, health, degeneration, and decay.
- **WDCH Dreams (September 2018)**: a commission for the LA Phil’s centennial. Trained on 587,763 images, 1,880 videos, 1,483 metadata files, and 17,773 audio files (44.5 TB in total), it was projected across the entire exterior of WDCH with 42 projectors (50K resolution) and 8-channel sound. A 12-minute piece in three parts (*Centennial Memories*, *Consciousness*, *Dream*) ran every 30 minutes over ten nights. Gehry’s 3D files let the studio map the building’s curves exactly. Indoors, a companion touchscreen installation let visitors explore the archive non-linearly.
- **Latent History (2019, Fotografiska Stockholm)**: about 150 years of Stockholm’s photographic archive reinterpreted with AI.
- **Machine Hallucinations: NYC (2019, Artechouse NYC)**: the first work in the series. Wikipedia describes 300 million photos of New York plus 113 million additional data points (subway sounds, radio, traffic noise); the artist’s site says “over 100 million photographic memories.” For privacy, only public images were used, and StyleGAN generated the “dreams.”
- **Quantum Memories (2020, NGV Triennial)**: more than 200 million public nature-related images combined with quantum computing research data from the Google AI Quantum team. Shown on the largest LED screen in NGV’s history, it won an iF Design Award in 2022.
- **Machine Hallucinations: Nature Dreams (2021, König Galerie Berlin)**: more than 300 million images of flowers, trees, mushrooms, landscapes, and clouds, then the largest raw nature dataset used for an artwork.
- **Casa Batlló residency (2021) and Living Architecture: Casa Batlló (May 2022)**: as Casa Batlló’s first artist in residence he created the six-sided cube room *In the Mind of Gaudí*. The following year, a 360-degree façade work trained on real-time climate data from sensors around the building and about one billion images (Gaudí’s sketches, the building’s archives, and more) drew 65,000 people.
- **Unsupervised - Machine Hallucinations - MoMA (opened 2022-11-19, ran until 2023-10-29)**: analyzed in detail in section 4.
- **Sense of Healing (2022 onward)**: commissioned for the LuisaViaRoma × UNICEF gala. An AI data sculpture made with Therme Art and MYND from the brain data of people in mental health treatment.
- **Living Paintings (February 2023, Jeffrey Deitch LA)**: his first major US solo exhibition, using Pacific wind forecasts, real-time weather, and images of California’s national parks.
- **Glacier Dreams (March 2023, Art Dubai)**: a work based on glacier data.

- **Machine Hallucinations: Sphere (September 2023 onward)**: the opening work for the Las Vegas Sphere, a 366-foot-tall building with a 580,000-square-foot LED shell. Part one used 1.1 million public images from the ISS, Hubble, and other sources; part two used 300 million national park images and wind sensor data from Las Vegas. NVIDIA A100s generated real-time AI animation, and it was then the largest AI artwork in the world. The artist called the premiere "one of the most Blade Runner moments ever."

3.3 In-house models based on nature and architecture (2024 to present)

- **Large Nature Model (LNM) and Living Archive: Nature (January 2024, WEF Davos)**: a generative AI model dedicated exclusively to nature, trained on images, sounds, and research from institutions such as the Natural History Museum in London and the Smithsonian. It is billed as "the world's first open-source generative AI model dedicated to nature."
- **Echoes of the Earth: Living Archive (2024-02-16 to 04-07, Serpentine North, London)**: the gallery walls were wrapped in imagery from data on flora, fauna, and fungi from more than 16 rainforest locations (including LiDAR and photogrammetry). More than 66,000 people visited in 47 days. It continued as his first solo exhibition in Asia at Futura Seoul in Gahoe-dong, Bukchon, Seoul (the exact opening date could not be confirmed in this research).
- **Large Nature Model: Living Art (March 2024, NVIDIA GTC, San Jose)** and **Large Nature Model: Coral (September 2024, UN Headquarters; Noor Riyadh 2024)**.
- **Winds of Yawanawá (2024)**: an NFT collection combining weather data from the Yawanawá village in the Amazon with artworks by young Yawanawá artists. It raised $3 million for the community.
- **AI Data Painting of the Keystone-Mast Collection (2024-09-21 to 2025-12-15, UCR California Museum of Photography)**: a reinterpretation of 250,000 stereoscopic glass-plate negatives and 100,000 prints.
- **Living Architecture: Gehry (2025-03-07 to 10-19, Guggenheim Museum Bilbao)**: the opening work of the Guggenheim's site-specific series *in situ*. The Large Architecture Model (LAM), an image diffusion model fine-tuned on Gehry's architectural language, generates an impossible building every minute, randomly varying location and time of day. Images are upscaled from a compact latent space to 16K, running continuously on Google Cloud. The work has six chapters: Data Universe, Data Plotting, Data Tunnel, Large Architecture Model, Machine Hallucination, and Dreams. Sound is by Kerim Karaoğlu.
- **Living Architecture: Biophilia (2025-05-10 to 11-23, Venice)**: installed during the 2025 Venice Architecture Biennale, where the artist also gave a keynote at the Biennale conference.

- **Living Memory: Messi, A Goal in Life (2025)**: Messi's header in the 2009 UEFA Champions League final turned into an AI data sculpture. Unveiled on June 11, 2025, it sold at Christie's for $1.87 million, with proceeds donated to educational initiatives in Latin America.
- **DATALAND and Machine Dreams: Rainforest (opened 2026-06-20, exhibition until 2027-01-31)**: covered in section 5.

4. Case study: *Unsupervised* (MoMA, 2022 to 2023)

Item	Details
Source data	138,151 collection records MoMA published on GitHub in 2016 (title, artist, medium, dimensions, date made, date acquired, classification)
Model	NVIDIA's StyleGAN2-ADA (adaptive discriminator augmentation), trained on subsets of the collection
Embeddings	1024 dimensions, then clustered by theme to capture semantic context
Exploration tool	Latent Space Browser, custom software the studio has developed since 2017
Visualization	Each data connection is drawn by an edge-detection algorithm and colored by the density of its previous and next latent coordinates, tracing the machine's "unconscious decisions" as lines
Fluid variation	*Fluid Dreams*: data pigment flows via a GPU-accelerated fluid solver and real-time ray-traced lighting
Data Universe variation	A "latent walk" built with UMAP over seven dimensions: x, y, z, r, g, b, and time. UMAP's inventor Leland McInnes praised it
Display	A roughly 24 × 24 foot media wall in MoMA's Gund Lobby, with trompe l'oeil depth at the screen edges
Reception	Extended four times after opening and ran for almost a year. Visitors stayed an average of 38 minutes. Acquired by MoMA in October 2023

The title (unsupervised learning) reflects the artist's point that "categorization requires a human perspective; without categories, the machine can find a new form." This reveals the core grammar of the work: **move the archive into a high-dimensional vector space, glide through that space like a camera, and render the movement as fluid and light.**

5. DATALAND (2026)

- **Location and scale**: The Grand LA (designed by Frank Gehry, 100 S Grand Ave), in the same cultural corridor as WDCH, The Broad, and MOCA. The public exhibition space is 25,000 square feet (Wikipedia gives 20,000 square feet). Gensler collaborated on the design, and Arup served as prime engineer and AV owner's representative.
- **Opening exhibition *Machine Dreams: Rainforest***: five galleries, including the Discovery Portal, the Data Pavilion with a 200-channel sound system, a gallery of cultural material shared with the permission of the Yawanawá community, and the Sanctuary, which visualizes aggregated audience interaction. It uses 1.5 billion pixels of high-resolution imagery, LiDAR and remote-sensing scans, more than 50,000 audio recordings, and real-time signals (humidity, light, soil moisture) from 16 rainforest locations.
- **Data.Link**: a wearable that pairs a wrist-worn Empatica device (spelled "Empathica" in the Artsy article) with a scent-diffusing neckband. The exhibition responds in real time to anonymized biometric and behavioral signals, and at the end each visitor receives a paper "data token" representing their visit, which they may keep or choose to delete.
- **Ethics and sustainability claims**: the museum says the model was trained on permission-based data from partners such as the Smithsonian, the Natural History Museum in London, Getty, and the Cornell Lab of Ornithology. It claims the LNM runs in a low-carbon Google Cloud region in Oregon on 87% carbon-free energy. The artist has said each visit uses roughly the energy of one iPhone charge. The press (Artsy) noted that these claims are likely to be closely scrutinized.
- **Additional platforms**: the open-access *Living Encyclopedia*, which gathers data on oceans, coral reefs, rainforests, and the atmosphere, and the DATALAND app for exchanging questions and answers with an AI inside the galleries.

6. Visual grammar: recurring vocabulary

Term	Meaning	Representative works
Data Pigment	Translating data values into the physical qualities of color and	Wind of Boston, Living Paintings

Term	Meaning	Representative works
	particles	
Data Universe	Spreading an archive through 3D like a nebula via dimensionality reduction (t-SNE, UMAP)	Archive Dreaming, Unsupervised: Data Universe, Living Architecture: Gehry chapter 1
Latent Walk	Continuous interpolation through the latent space of a GAN or diffusion model, so forms transform endlessly	Machine Hallucinations throughout
Fluid Dreams	Pigment flowing and swirling via a GPU fluid solver	Unsupervised Fluid Dreams, Sphere
Data Sculpture	A trompe l'oeil volume that seems to have depth inside its frame	Unsupervised, Coral Dreams, Messi
Living Architecture	Projection or LED skins fitted to a building's actual 3D form	WDCH Dreams, Casa Batlló, Sphere
Infinity Room	An immersive room in which every surface is a projection surface	Infinity Room, In the Mind of Gaudí
Data Tunnel	A first-person flight through latent space	Living Architecture: Gehry chapter 3

A detailed technical breakdown is in `03-technique-analysis.md`.

7. Criticism and debate

1. **The spectacle critique**: Jerry Saltz (New York/Vulture) called *Unsupervised* "a massive techno lava lamp," "a half-million-dollar screensaver," and a digital version of the "Zombie Formalism of the 2010s." His point is that it is a derivative that fails to transcend its source material and "takes away art's otherness." The two later had a public online spat (reported by Artnet).
2. **Art-historical precedent**: in the same piece Saltz cited the digital transformations of Jason Salavon and Jeremy Blake in the early 2000s, saying *Unsupervised* "adds nothing but scale and expense."
3. **The defense**: Feral File and others see *Unsupervised* as a "historical moment" where three early-2020s currents met: immersive installation, NFTs, and AI. The 38-minute dwell time and MoMA's acquisition are often cited as evidence of its public appeal.

4. **Data ethics and environmental cost**: early on he described using "publicly available images," but since 2024 he has foregrounded "permission-based" data and low-carbon computing. This can be read as a response to public demands about data provenance and carbon cost, and at DATALAND these claims have become subject to scrutiny.
5. **NFTs and the market**: since 2020 he has released more than a dozen collections and over 11,000 tokens, raising about $10 million for charity in total. This is entangled with criticism about the boundary between art and the market.

8. Awards (selected)

TIME100 Impact Award (2025), UCLA Edward A. Dickson Alumnus of the Year (2024), iF Gold Award (Quantum Memories, 2022), Lumen Prize Gold (2019), Lorenzo il Magnifico Lifetime Achievement Award for New Media Art, Microsoft Research Best Vision Award.

9. Implications for implementation

Anadol's work is a pipeline of **(1) large-scale data collection, (2) training and embedding, (3) latent space exploration, (4) real-time rendering, and (5) architectural display**. Of these, what a web browser and three.js can reproduce is the visual grammar of steps (3) to (5). Rather than reproducing the trained models themselves, it is realistic to implement the following as GPU shaders:

- Data Universe, spreading high-dimensional data into a 3D point cloud
- Data Pigment fluids, advecting particles with curl noise or real wind data
- Latent Walk morphs that interpolate endlessly between forms
- Data Sculpture and Infinity Room, with depth inside a frame

The concrete step-by-step plan is in `05-threejs-plan.md`. This project is a technique study and homage; it does not copy or use the artist's original videos, images, or datasets.

Refik Anadol Timeline

Research cutoff: 2026-10-15. The full structured list is in `../../data/works.json` and `../../data/works.csv`.

Overview

```
timeline
    title Refik Anadol key timeline (1985 to 2026)
    1985 : Born in Istanbul
    2008 : First outdoor projection, begins using the term "data painting"
    2010 : Quadrature (santralistanbul)
    2011 : Augmented Structures v1.0 (Istanbul Biennial)
    2012 : Enters UCLA DMA, develops Infinity Room
    2014 : Visions of America – Amériques (WDCH) : Refik Anadol Studio founded
    2015 : Infinity Room unveiled (Istanbul Biennial)
    2016 : First Google AMI residency
    2017 : Wind of Boston : Archive Dreaming (SALT, 1.7 million documents)
    2018 : Melting Memories : WDCH Dreams (LA Phil centennial)
    2019 : Machine Hallucinations – NYC : Latent History (Fotografiska) : Lumen Prize Gold
    2020 : Quantum Memories (NGV) : First NFT
    2021 : Machine Hallucinations – Nature Dreams : Casa Batlló residency
    2022 : Living Architecture – Casa Batlló : Unsupervised (MoMA opening)
    2023 : Living Paintings (Jeffrey Deitch) : Unsupervised enters MoMA collection : Machine
Hallucinations – Sphere
    2024 : Large Nature Model (Davos) : Echoes of the Earth (Serpentine) : LNM Coral (UN)
    2025 : Living Architecture – Gehry (Guggenheim Bilbao) : Biophilia (Venice) : Messi data
sculpture : TIME100 Impact Award
    2026 : DATALAND opens (June 20) : Machine Dreams – Rainforest
```

Evolution of technique

```
flowchart LR
    A["2008 to 2015<br/>Projection mapping<br/>architecture = canvas"] --> B["2016 to 2018<br/>t-
SNE, DCGAN<br/>archive = universe"]
    B --> C["2019 to 2023<br/>StyleGAN2-ADA, UMAP<br/>GPU fluid solver<br/>latent space = dream"]
    C --> D["2024 onward<br/>In-house models LNM, LAM<br/>fine-tuned diffusion<br/>permission-
based data"]
    D --> E["2026<br/>DATALAND<br/>wearables + live ecological data"]
```

Major exhibition periods, 2022 to 2026

```
gantt
    dateFormat YYYY-MM-DD
    axisFormat %Y-%m
```

```
    title Major exhibition periods (confirmed dates only)
    section MoMA
    Unsupervised                :2022-11-19, 2023-10-29
    section Sphere
    Machine Hallucinations Sphere :2023-09-01, 120d
    section London
    Echoes of the Earth (Serpentine) :2024-02-16, 2024-04-07
    section Riverside
    Keystone-Mast AI Data Painting :2024-09-21, 2025-12-15
    section Bilbao
    Living Architecture Gehry   :2025-03-07, 2025-10-19
    section Venice
    Living Architecture Biophilia :2025-05-10, 2025-11-23
    section LA
    DATALAND Machine Dreams Rainforest :2026-06-20, 2027-01-31
```

Note: the Sphere run is an approximation based on Wikipedia's description of a "four-month installation". The artist's site lists it as "ongoing since 2023-09-01".

Technique Breakdown: Refik Anadol's Data Aesthetics Pipeline

This document reconstructs the techniques described in public materials (work descriptions on the artist's site, museum texts, press articles). The studio's internal code is not public, so "publicly stated techniques" and "reasonable inferences" are kept separate.

1. Overall pipeline

```
flowchart TD
    subgraph S1["1. Collection"]
        D1["Archive images<br/>(MoMA 138,151 records, SALT 1.7 million)"]
        D2["Environmental sensors<br/>(wind speed, humidity, soil moisture)"]
        D3["Biosignals<br/>(EEG, wearables)"]
    end
    subgraph S2["2. Training and embedding"]
        M1["GAN training<br/>DCGAN, PGAN, StyleGAN2-ADA"]
        M2["Diffusion fine-tuning<br/>LNM, LAM"]
```

```
        M3["Dimensionality reduction<br/>t-SNE, UMAP"]
    end
    subgraph S3["3. Latent space exploration"]
        L1["Latent Space Browser<br/>(in-house, since 2017)"]
        L2["Latent walk paths<br/>(interpolation, cluster tours)"]
    end
    subgraph S4["4. Real-time rendering"]
        R1["GPU fluid solver<br/>data pigment"]
        R2["Point cloud<br/>Data Universe"]
        R3["Real-time ray-traced lighting"]
    end
    subgraph S5["5. Architectural display"]
        X1["Projection mapping<br/>from building 3D models"]
        X2["LED walls, spherical shells"]
        X3["Multichannel sound"]
    end
    D1 --> M1 & M3
    D1 --> M2
    D2 --> R1
    D3 --> R1
    M1 --> L1
    M2 --> L1
    M3 --> R2
    L1 --> L2 --> R1 & R2
    R1 & R2 & R3 --> X1 & X2
    X3 -.-> X1
```

2. Stage by stage

2.1 Collection: from “public data” to “permission-based data”

Period	Data policy	Examples
2017 to 2023	Public archives, public images from social media and search engines	Machine Hallucinations NYC (public photos), Unsupervised (MoMA’s public metadata on GitHub)
2024 onward	Permission-based data through institutional partnerships, on-site LiDAR and photogrammetry	LNM (Smithsonian, Natural History Museum London, Getty, Cornell Lab of Ornithology)

Publicly stated: the scale of collection (hundreds of millions of images) and the categories of sources. **Inferred:** preprocessing such as deduplication, quality filtering and resolution normalization is standard for large-scale GAN training, so it was likely applied.

2.2 Training and embedding

- **GAN family (2016 to 2023):** DCGAN (Archive Dreaming), PGAN, StyleGAN, StyleGAN2-ADA (Unsupervised). StyleGAN2-ADA adds adaptive data augmentation to the discriminator, reducing overfitting even on relatively small datasets. It suits curated archives of a few hundred thousand images or fewer, such as the MoMA collection.
- **1024-dimensional embeddings (Unsupervised):** images were represented as high-dimensional vectors and then clustered by theme.
- **Dimensionality reduction:** t-SNE (Archive Dreaming, WDCH Dreams) and UMAP (Unsupervised: Data Universe, Stockholm archive studies). The output becomes x, y, z coordinates, and adding color (r, g, b) and time makes a "seven-dimensional" work.
- **Diffusion models (2024 onward):** LAM is an image diffusion model fine-tuned on Gehry's architectural language. It generates in a compressed latent space, upscales to 16K with in-house upscaling models, and makes a new image every minute while randomly changing location and time of day.

2.3 Latent space exploration

- **Latent Space Browser:** an in-house tool developed since 2017 for exploring a trained latent space and choosing "hallucinations". According to the Coral Dreams description, the artist personally selects particular hallucinations from "countless serendipitous allusions to coral". It is not fully automated; this is **where the artist's curation intervenes**.
- **Latent walk:** interpolating the latent vector z continuously (for example with spherical linear interpolation, slerp) makes the generated image transform without breaks. This is the source of the "melting into something else" motion characteristic of Machine Hallucinations.
- **Edge and density coloring (Unsupervised):** data connections are drawn as lines with an edge-detection algorithm and colored by the density of the previous and next latent coordinates. The path of movement itself becomes the drawing.

2.4 Real-time rendering

- **GPU fluid solver:** the Unsupervised Fluid Dreams description explicitly mentions "fluid solver algorithms accelerated by GPU computation and real-time ray-traced lighting". Colors drawn from

the latent space are injected into a fluid field like pigment and flow.

- **Real-time generation:** Sphere produced real-time AI animation on NVIDIA A100s, and LAM and LNM run continuously on Google Cloud.
- **Inferred tools:** a collaborator's portfolio confirms Notch (Bilbao art direction). TouchDesigner, Houdini and Unreal-family tools are also plausible by industry practice, but this research could not confirm them officially.

2.5 Architectural display

- **Original building 3D models:** WDCH Dreams mapped the curved surfaces precisely using 3D files provided by Gehry, with 42 projectors, 50K resolution and 8-channel sound.
- **Trompe-l'œil frame:** the 24 × 24 ft LED wall at MoMA used a depth illusion in which forms seem to burst past the screen edges (Saltz also mentions the "trompe l'oeil edges").
- **Immersive rooms:** Infinity Room and In the Mind of Gaudí are cube rooms projected on all six sides.
- **Multisensory expansion:** DATALAND widened the senses with 200-channel sound, scent-diffusing neckbands and wearable biosignals.

3. Mapping to the web (three.js)

Anadol technique	Web counterpart	Difficulty	Notes
Data pigment fluid	GPU compute particles + curl noise or a 2D fluid solver (Stable Fluids)	Medium	Hundreds of thousands of particles in real time with WebGPU compute shaders
Data Universe	Precomputed UMAP coordinates (JSON) + `Points` or instancing	Low	Real embeddings generated offline in Python
Latent walk	Particle position interpolation between target shapes + noise	Medium	A "shape space" stands in as a metaphor for the latent space instead of a GAN
Trompe-l'œil data sculpture	Fixed camera, box frame, depth fog, bloom	Low	Fixed aspect ratio

Anadol technique	Web counterpart	Difficulty	Notes
Infinity Room	Shader materials on the six inner faces of a cube	Low	First-person camera
Data tunnel	Recycled particle tunnel following the camera	Low	
Real-time environmental data	Inject public weather API (wind) data into the fluid field	Medium	If an API key is needed, route through a server-side proxy
Biometric response	Microphone (Web Audio), pointer movement	Low	No personal data is collected or stored

Scene Catalog: 10 Themes × 10 = 100

Principle: each scene covers **one concept**, and no phenomenon or subject appears twice. Scenes that use a similar algorithm (for example, branching) count as separate concepts when their subject and scale differ, and even then their visual treatment is designed to look different. Numbering follows the theme order (theme number × 10 + position).

Theme	File	Scope
1 Machine Dreams	`themes/machine.js`	Anadol's grammar of data and AI
2 Atmosphere	`themes/atmosphere.js`	Weather phenomena (macro)
3 Ocean & Water	`themes/ocean.js`	Oceans and waterways
4 Forest & Life	`themes/life.js`	Plants, animals, and ecology
5 Earth & Geology	`themes/earth.js`	Landforms and geological processes
6 Cosmos	`themes/cosmos.js`	Celestial bodies and cosmic structure (ultra-macro)
7 Micro	`themes/micro.js`	Atoms, molecules, and cells
8 City	`themes/city.js`	Urban infrastructure and life
9 Countryside	`themes/countryside.js`	Rural landscapes and labor
10 Physics & Math	`themes/physics.js`	Laws and mathematical structures

1 Machine Dreams

#	ko	en	Concept and visualization
1	데이터 안료	Data Pigment	Particles carried by a curl-noise fluid (original)
2	데이터 유니버스	Data Universe	An archive grid regrouping into an embedding nebula (original)
3	잠재 산책	Latent Walk	Interpolation through a space of shapes (original)
4	무한의 방	Infinity Room	A room projected on all six faces (original)
5	데이터 터널	Data Tunnel	A first-person flight through latent space (original)
6	녹아내리는 기억	Melting Memories	EEG ribbons melting (original)
7	양자 기억	Quantum Memories	A probability cloud and measured voxels (original)
8	신경망	Neural Network	Activations flowing along connections between layered nodes
9	어텐션 지도	Attention Map	Attention arches between tokens arranged in a ring
10	경사 하강	Gradient Descent	Walkers rolling into the valleys of a loss landscape

2 Atmosphere

#	ko	en	Concept and visualization
11	바람 회화	Wind Painting	2D wind flow (original)
12	오로라	Aurora	Curtains of light hanging in the sky, rippling vertical bands
13	번개	Lightning	A branching discharge path that flashes
14	토네이도	Tornado	A rotating funnel column with entrained debris
15	허리케인	Hurricane	Spiral cloud bands and an eye seen from above
16	적운	Cumulus	Billowing masses of cloud
17	몬순	Monsoon	Rain pouring down at a slant
18	눈 내림	Snowfall	Snowflakes fluttering down
19	제트 기류	Jet Stream	A meandering band of Rossby waves wrapped around the globe
20	안개	Valley Fog	A low layer of fog filling and flowing through a valley

3 Ocean & Water

#	ko	en	Concept and visualization
21	산호의 꿈	Coral Dreams	A branching coral data sculpture (original)
22	너울	Ocean Swell	A sea surface moving with Gerstner waves

#	ko	en	Concept and visualization
23	소용돌이	Maelstrom	A rotating water surface drawn into its center
24	폭포	Waterfall	A falling stream of water and its spray
25	해파리	Jellyfish	Pulsing bells with trailing tentacles
26	발광 플랑크톤	Bioluminescence	Blue plankton that glow when a wave touches them
27	물고기 떼	Fish School	A school of fish milling in a rotating vortex
28	빙산	Iceberg	A small part above the waterline and a huge body below
29	열수 분출공	Hydrothermal Vent	A column of black smoke rising from a deep-sea chimney
30	강 삼각주	River Delta	Branching channels spreading into the sea, seen from above

4 Forest & Life

#	ko	en	Concept and visualization
31	기계의 꿈: 열대우림	Machine Dreams: Rainforest	Vines and spores (original)
32	나무의 성장	Tree Growth	A 3D tree whose branches grow out from the trunk in turn

#	ko	en	Concept and visualization
33	균사 네트워크	Mycelium	Hyphae spreading underground, with nutrient signals
34	반딧불이	Fireflies	Fireflies whose flashing gradually synchronizes
35	철새의 이동	Bird Migration	Flocks moving along great-circle routes over the globe
36	벌집	Beehive	Hexagonal comb cells with bees coming and going
37	단풍 낙엽	Autumn Leaves	Leaves tumbling as they fall
38	민들레 홀씨	Dandelion Seeds	Seeds scattering on their pappus
39	거미줄	Spider Web	An orb web woven from radial threads and a spiral thread
40	개미 행렬	Ant Trails	Ants traveling back and forth along pheromone trails

5 Earth & Geology

#	ko	en	Concept and visualization
41	화산 분화	Volcano	Ejecta shooting up from the crater and falling back
42	판 구조	Tectonic Drift	Plates pulling apart with a ridge rising between them

#	ko	en	Concept and visualization
43	사구	Sand Dunes	Crescent sand dunes migrating with the wind
44	협곡 지층	Canyon Strata	Colored bands of strata in eroded canyon walls
45	정동	Geode	Crystals lining the inside of a hollow rock
46	지진파	Seismic Waves	P waves and S waves spreading through a cross-section of the Earth
47	종유굴	Stalactite Cave	Stalactites growing from the ceiling, with falling drips
48	산맥	Mountain Range	A low terrain flight over the ridgelines
49	간헐천	Geyser	A column of steam and water erupting at intervals
50	소금 사막	Salt Flat	A salt plain cracked into hexagons

6 Cosmos

#	ko	en	Concept and visualization
51	나선 은하	Spiral Galaxy	Density-wave spiral arms in differential rotation
52	블랙홀	Black Hole	An accretion disk and a photon ring
53	성운	Nebula	Gas clouds where stars are born

#	ko	en	Concept and visualization
54	태양 코로나	Solar Corona	Magnetic loops and flares on the Sun’s surface
55	행성의 고리	Planetary Rings	A thin ring of particles orbiting a planet
56	초신성	Supernova	A shock-wave shell expanding outward
57	혜성	Comet	A tail streaming away from the Sun
58	우주 거대 구조	Cosmic Web	Galaxy filaments and voids
59	펄서	Pulsar	Two beams sweeping around as the star rotates
60	태양계	Orrery	Planets on Keplerian orbits

7 Micro

#	ko	en	Concept and visualization
61	원자 오비탈	Atomic Orbitals	s, p, and d electron probability clouds
62	DNA 이중 나선	DNA Helix	Two rotating strands and their base pairs
63	세포 분열	Mitosis	A cell and its chromosomes dividing in two
64	바이러스 캡시드	Virus Capsid	An icosahedral protein shell
65	혈류	Blood Flow	Red blood cells flowing through a vessel

#	ko	en	Concept and visualization
66	단백질 접힘	Protein Folding	An unfolded chain folding into a structure
67	브라운 운동	Brownian Motion	Particles jittering from molecular collisions
68	세균 군집	Bacterial Colony	Colonies spreading across a culture dish
69	규조류	Diatoms	Radially symmetric silica shells
70	눈 결정	Snowflake	A hexagonal dendritic crystal growing

8 City

#	ko	en	Concept and visualization
71	스피어	Sphere	A spherical LED shell (original)
72	살아있는 건축	Living Architecture	Projection on a curved facade (original)
73	고속도로 빛의 궤적	Highway Trails	Streams of vehicle headlights and taillights
74	지하철 노선망	Metro Network	Trains running along their lines
75	마천루 창문	Skyline Windows	High-rise windows switching on and off
76	야경 항공뷰	City Grid at Night	A grid of streetlights seen from above
77	스크램블 교차로	Scramble Crossing	Pedestrians crossing in every direction at once

#	ko	en	Concept and visualization
78	불꽃놀이	Fireworks	Fireworks bursting over the city
79	현수교	Suspension Bridge	Main cables, hangers, and traffic on the deck
80	전파	Radio Signals	Concentric radio waves spreading from base stations

9 Countryside

#	ko	en	Concept and visualization
81	밀밭	Wheat Field	Ears of wheat rippling in the wind
82	다랑논	Rice Terraces	Flooded, terraced rice paddies
83	풍력 발전기	Wind Turbines	Blades turning on a hill
84	들판 조각보	Farmland Patchwork	Fields and furrows seen from above
85	미스터리 서클	Crop Circle	A geometric pattern drawn into a wheat field
86	연못의 빗방울	Pond Ripples	Concentric ripples spreading across a pond
87	굴뚝 연기	Chimney Smoke	Smoke rising from the village rooftops
88	해바라기 밭	Sunflower Field	Fibonacci flower heads that turn to follow the Sun
89	방목지	Grazing Pasture	Livestock wandering slowly across a meadow

#	ko	en	Concept and visualization
90	물레방아	Waterwheel	A wheel turning as it lifts water

10 Physics & Math

#	ko	en	Concept and visualization
91	로렌츠 끌개	Lorenz Attractor	A butterfly-shaped chaotic trajectory
92	진자 파동	Pendulum Wave	Waves formed by pendulums of different lengths
93	리사주 곡선	Lissajous	Curves set by the frequency ratio of two oscillations
94	자기력선	Magnetic Field	Field lines around a dipole
95	클라드니 도형	Chladni Plate	Sand gathering on the nodal lines of a vibrating plate
96	파동 간섭	Wave Interference	Constructive and destructive patterns from two sources
97	뫼비우스 띠	Möbius Strip	The topology of a strip with only one side
98	망델브로 집합	Mandelbrot Set	A fractal terrain whose height is the escape speed
99	보로노이 분할	Voronoi Shatter	Space breaking into cells around seed points
100	프리즘 분산	Prism Dispersion	White light splitting into a rainbow spectrum

Sources

Research date: 2026-10-15. All URLs were checked on this date via Tavily search and extraction.

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Conflicts between sources

Item	Value A	Value B
Machine Hallucinations: NYC data	300 million photos + 113 million data points (Wikipedia)	100 million+ photos (artist site)
DATALAND floor area	25,000 sq ft of public exhibition space (Artsy, Getty)	20,000 sq ft (Wikipedia)

Item	Value A	Value B
Living Architecture: Gehry end date	2025-10-19 (designboom, events page text)	2025-10-12 (work page metadata)